

15. Let $i = \sqrt{-1}$ and let k represent a real number. If $7 + ki^3 + x = 4 + 6i^2 + 3i$ is solved for x , then $x = -9 - 7i$. Find the value of k .
16. Let n represent an integer such that $0 < n < 6$. Two of the roots for x of the cubic equation $x^3 + nx^2 - (n+1)x + k = 0$ are additive inverses. Find the sum of all possible distinct values of k .
17. Katie randomly places 3 identical math, 2 identical science, and 2 identical English books on a shelf. Find the probability that the 3 math books are together on the shelf. Express your answer as a common fraction reduced to lowest terms.
18. Given that $\sin(A) = -\frac{2}{3}$ and $\pi < A < \frac{3\pi}{2}$ (where A is measured in radians). Then $\sin(2A)$ can be expressed in simplest radical form as $\frac{k\sqrt{w}}{p}$ where k and p are relatively prime integers and $p > 0$. Find the value of $(k + w + p)$.
19. Let $i = \sqrt{-1}$ and let $(x-1-2i)$, $(x-1+2i)$, and $(x-5)$ be factors of the polynomial $x^3 + kx^2 + wx + p$ where k , w , and p represent integers. Find the value of $(k + p)$.
20. In $\triangle ABC$, let $AB = c$, $AC = b$, and $BC = a$. $\angle BAC = 60^\circ$, and $(a+b+c)(a+b-c) = \frac{49}{19}ab$. If the lengths of all sides of the triangle are integers, find the smallest possible value of c .